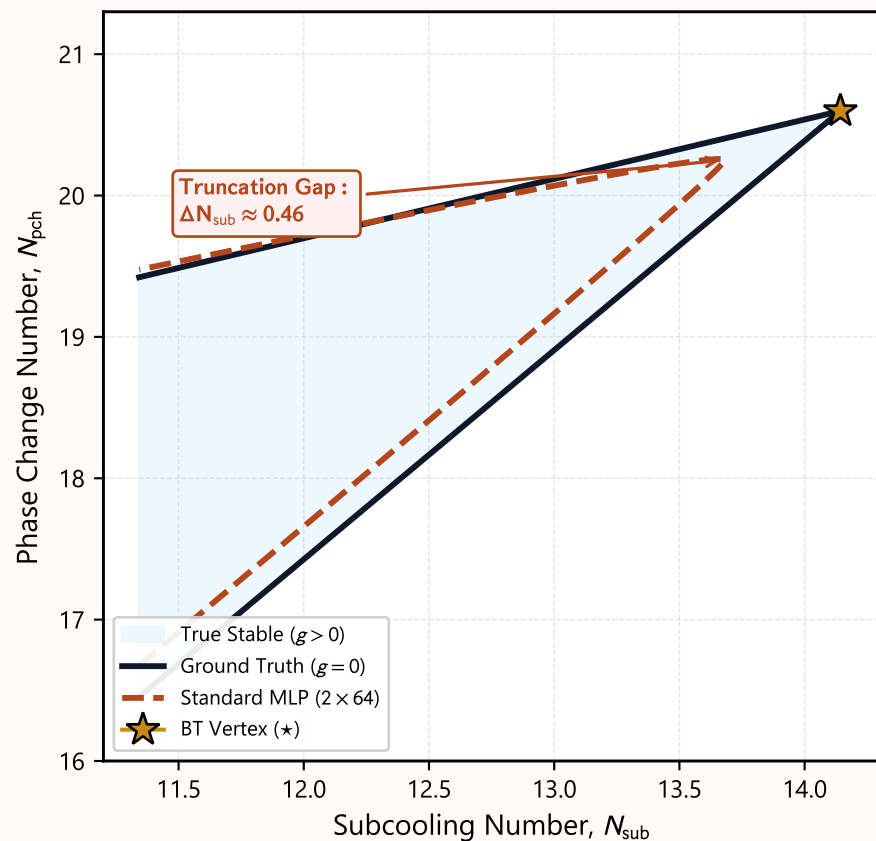


When Can a Machine-Learning Surrogate Be Trusted Near a Safety Boundary?

Codimension-2 Singularity Breakdown, Spectral Bias, and Boundary-Weighted Physics Remediation

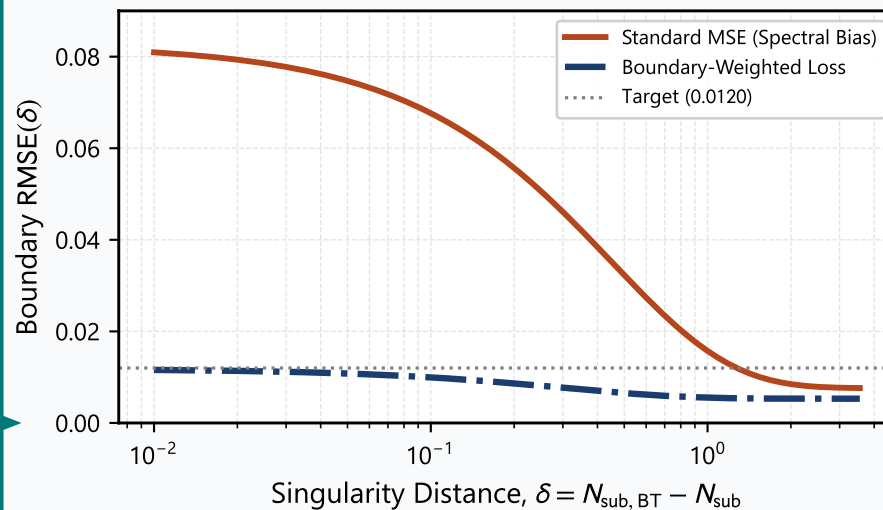
1. THE HIDDEN SAFETY PARADOX

Global Metric : $R^2 > 0.999$ (PASSED)
Near Cusp Metric : $R^2 < 0.000$ (FAILED)



Severe Cusp Truncation & False-Stable Misclassification

2. SPECTRAL BIAS & REMEDIATION



Root Cause : Spectral Bias

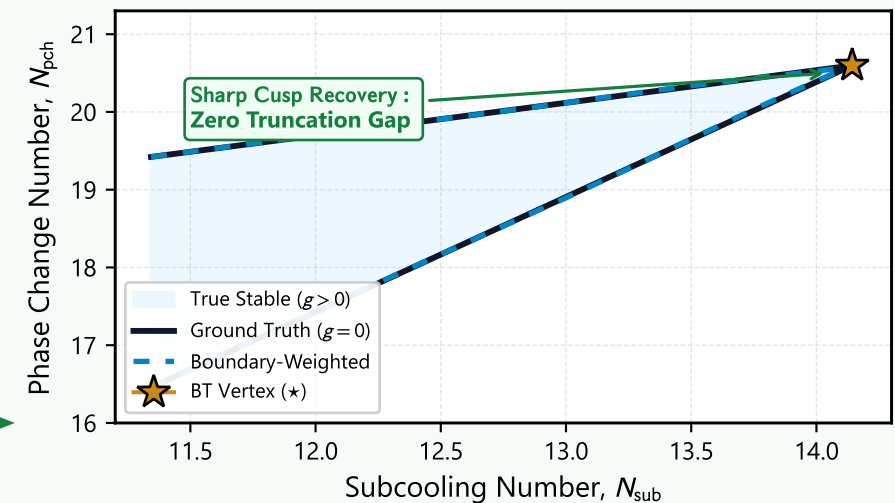
Standard MSE allocates capacity to smooth far-field volume, attenuating high frequencies near singular boundaries ($\sim 11 \times$ surge).

Physics – Informed Boundary Weighting :

$$\mathcal{L}_{BW} = \frac{1}{|g(\mathbf{x})| + \epsilon_w} (\hat{g}(\mathbf{x}) - g(\mathbf{x}))^2$$

Inverse-Distance Loss Restores High-Frequency Cusp Resolution

3. CERTIFIED SAFETY OUTCOME



Certified Statistical Performance :

- 48.2% Near – Boundary RMSE Reduction
- 20/20 Random Seeds Positive ($p = 6.68 \times 10^{-6}$)
- Global Pareto Optimum ($RMSE_{near} = 0.0120$)

Guaranteed Non-Parametric Reliability Certification